OBJECTIVE:

Designing a universal shift register using flip-flops and multiplexers.

COMPONENT ANALYSIS:

Name		Quantity
i) Trainer board	–	1 piece
ii) IC extractor	–	1 piece
iii) IC (Dual 4×1 MUX) 74153	–	2 pieces
iv) IC (Dual D-FF) 7474	–	2 pieces
v) Magnifier	–	1 piece
vi) Handbook	–	1 piece
vii) Wires		

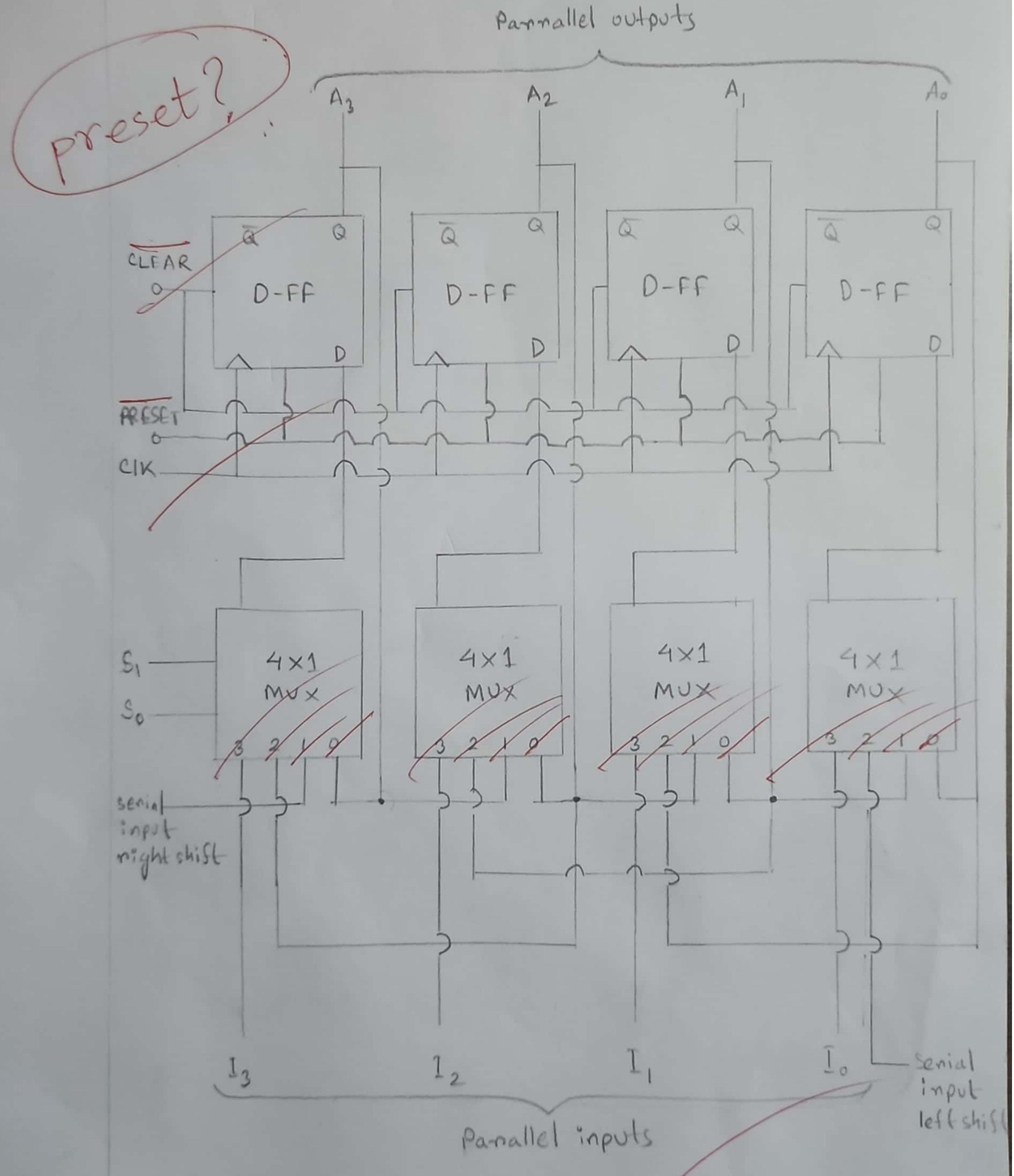
PROBLEM SPECIFICATION:

Design and implement a 4-bit universal shift register.

EXCITATION TABLE:

Input					Output				
CLEAR	Clock	S_0	S_1	P_i	A_3	A_2	A_1	A_0	Mode
0	×	×	×	×	0	0	0	0	Asynchronous clear
1	×	0	0	×	A_3	A_2	A_1	A_0	Hold
1	↑	0	1	A_{i+1}	I_R	A_3	A_2	A_1	Shift right
1	↑	1	0	A_{i+1}	A_2	A_1	A_0	I_L	Shift left
1	↑	1	1	A_i	I_3	I_2	I_1	I_0	Parallel load

DESIGN:


preset?
Parrallel outputs
A_3
A_2
A_1
A_0
$\overline{CLEAR}$
$\overline{Q}$
Q
D-FF
D
$\overline{PRESET}$
ClK
S_1
S_0
4x1
MUX
3
2
1
0
serial input right shift
I_3
I_2
I_1
I_0
serial input left shift
Parallel inputs

CIRCUIT DIAGRAM: